

A decorative graphic on the left side of the slide, consisting of a network of white and light blue lines and circles, resembling a circuit board or a stylized tree structure.

RECTILINEAR MOTION

CHAPTER 15 (PART 1)



Objectives

-  List the characteristics of a uniform rectilinear motion (URM).
-  Determine the characteristics of a motion from its graph.

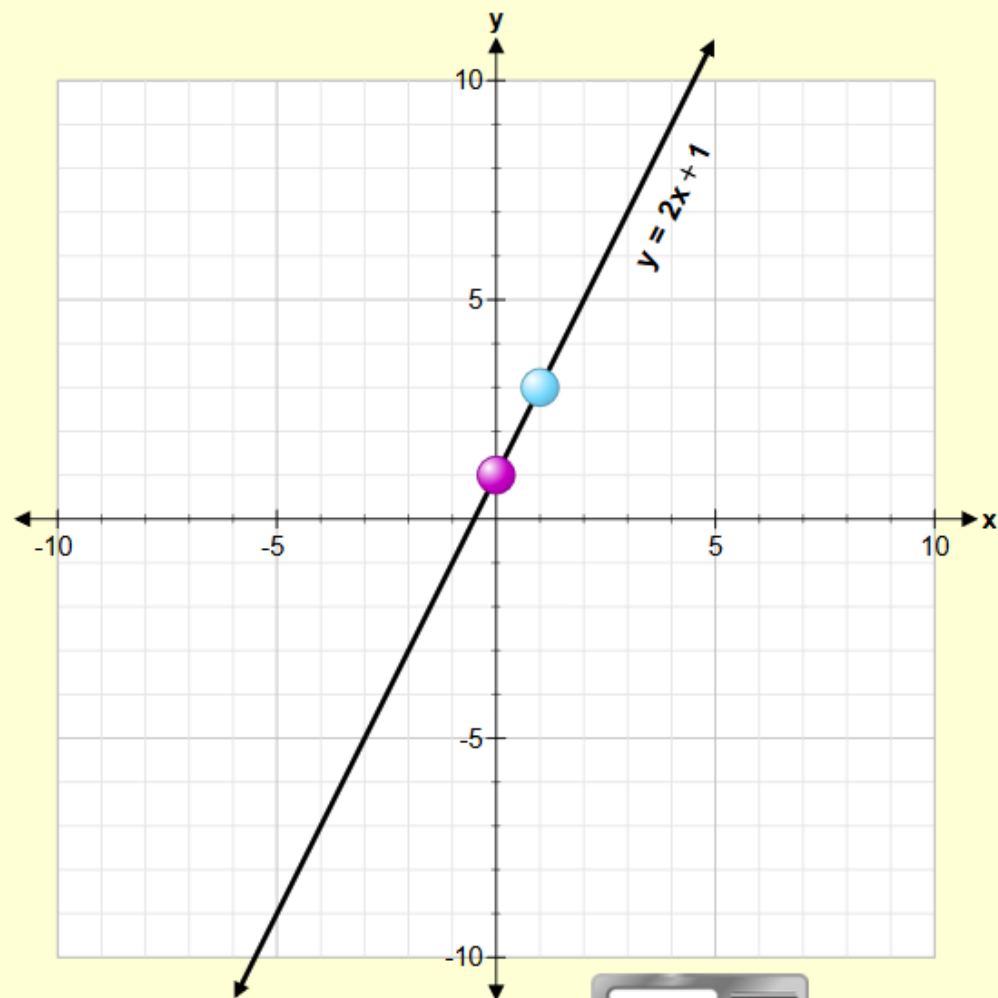
Remember

The general form of a straight line equation is:

$$y = a \cdot x + b$$

Slope of the
straight line

Y-intercept
(for $x = 0 \rightarrow y = b$)



(?, ?)

(?, ?)

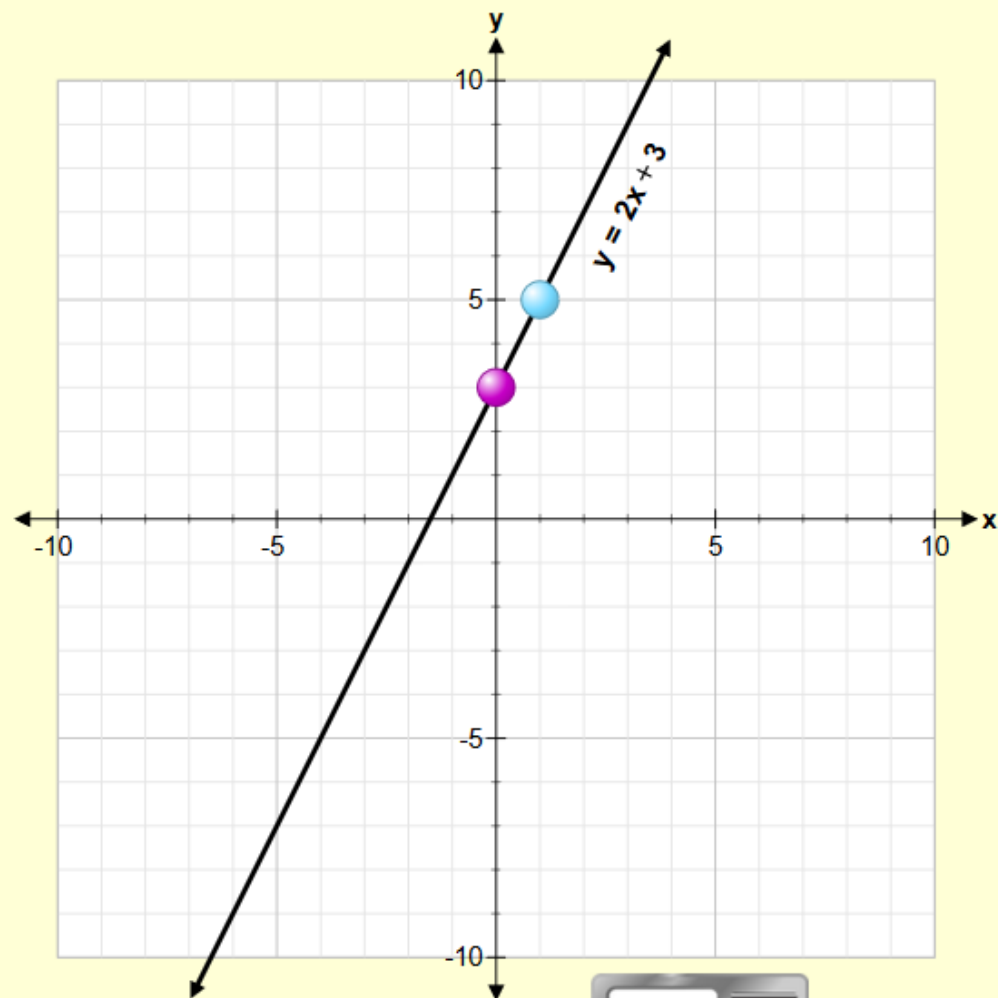
$y = mx + b$

$y = \frac{2}{1}x + 1$

Save Line Erase Lines

- ☐ Slope
- ☐ $y = x$
- ☐ $y = -x$
- ☐ Hide lines
- ☐ Hide grid





(?, ?)

(?, ?)

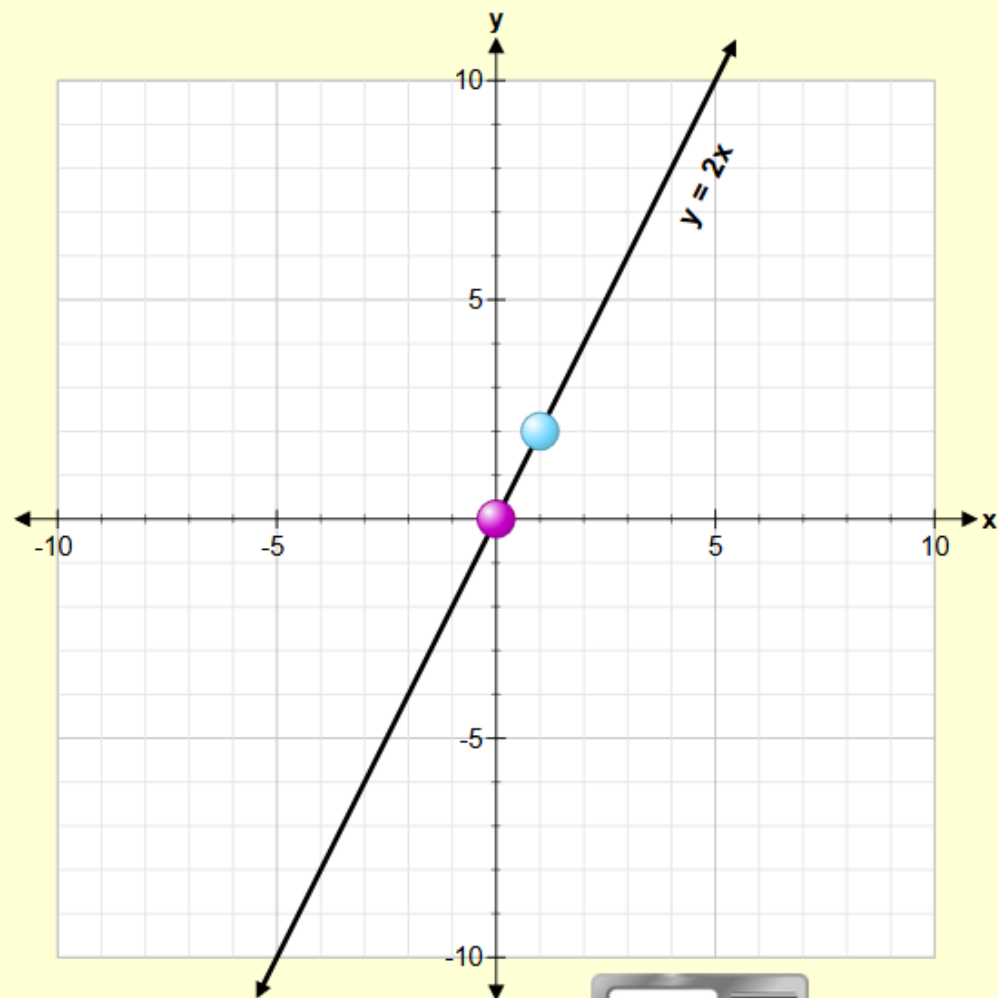
$y = mx + b$

$y = \frac{2}{1}x + 3$

Save Line Erase Lines

- ☐ Slope
- ☐ $y = x$
- ☐ $y = -x$
- ☐ Hide lines
- ☐ Hide grid





(?, ?)

(?, ?)

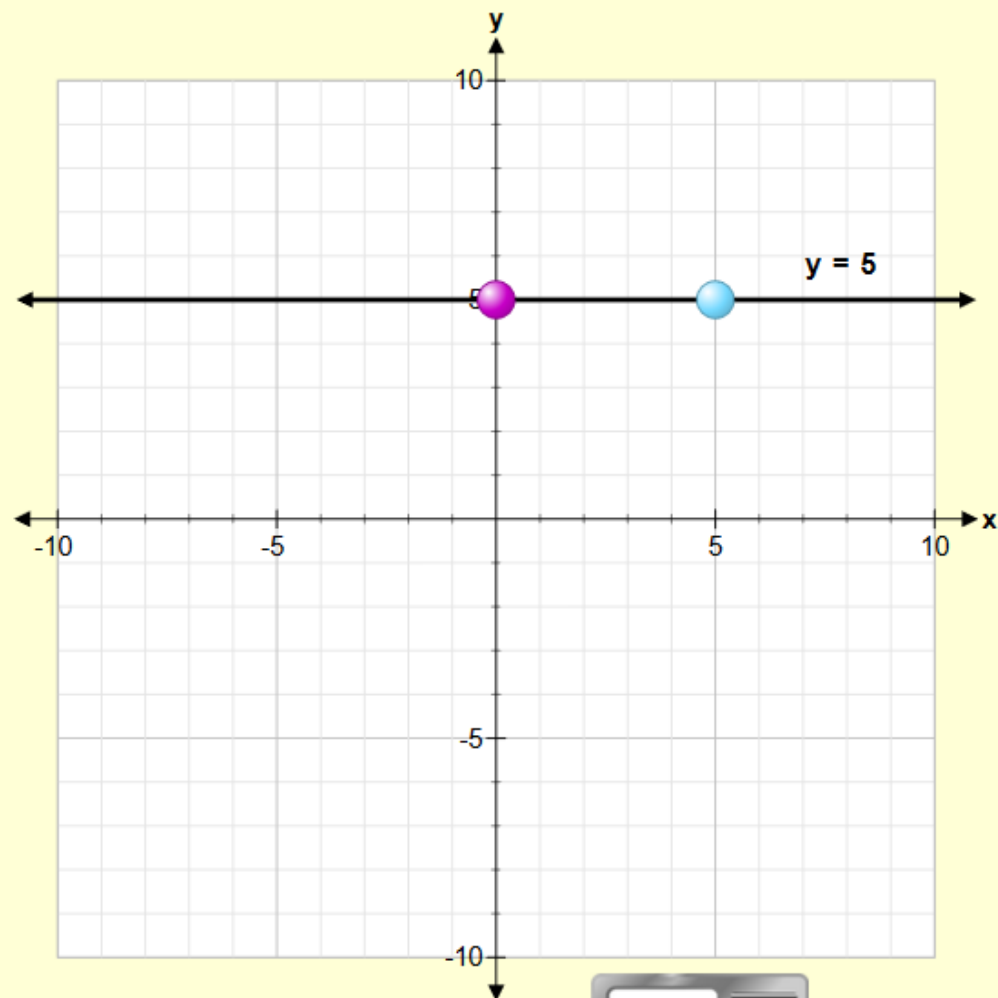
- $y = mx + b$

$$y = \frac{2}{1}x + 0$$

Save Line Erase Lines

- ☐ Slope
- ☐ $y = x$
- ☐ $y = -x$
- ☐ Hide lines
- ☐ Hide grid





(?, ?)

(?, ?)

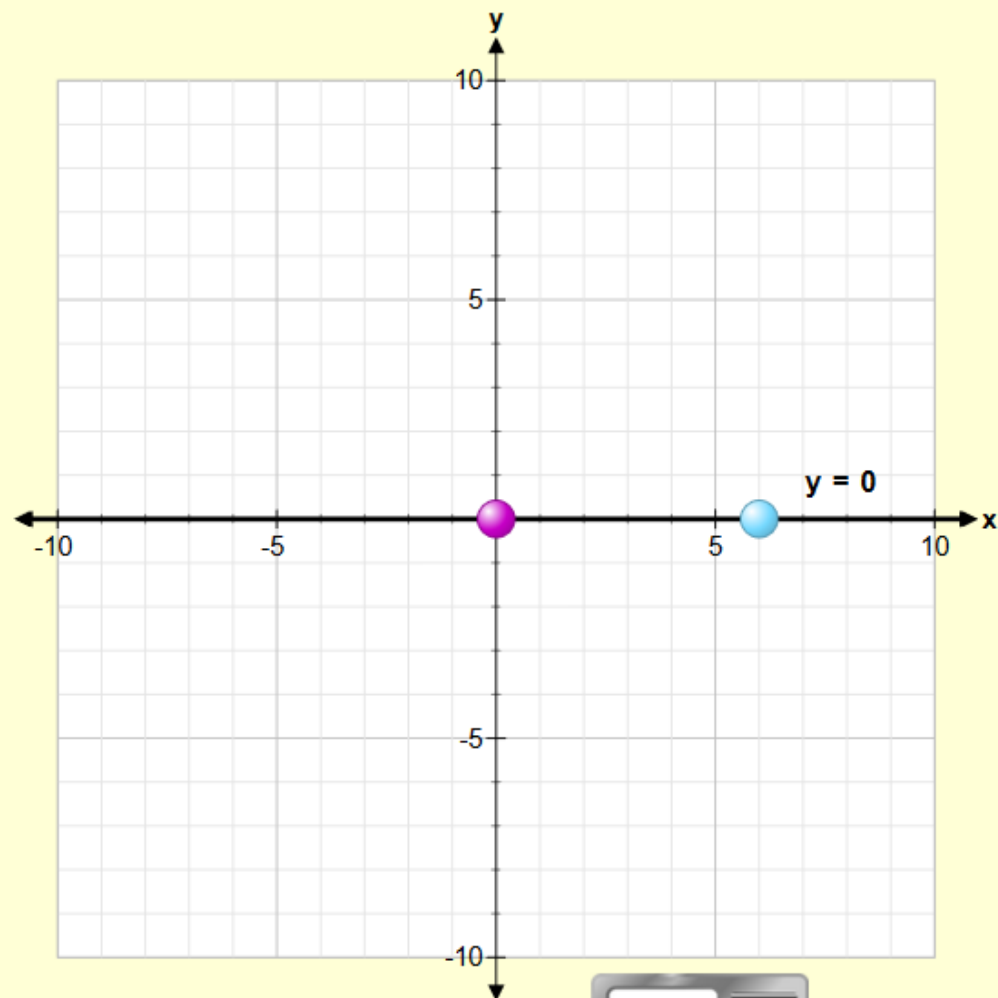
$y = mx + b$

$y = \frac{0}{5}x + 5$

Save Line Erase Lines

- ☐ Slope
- ☐ $y = x$
- ☐ $y = -x$
- ☐ Hide lines
- ☐ Hide grid





(?, ?)

(?, ?)

- $y = mx + b$

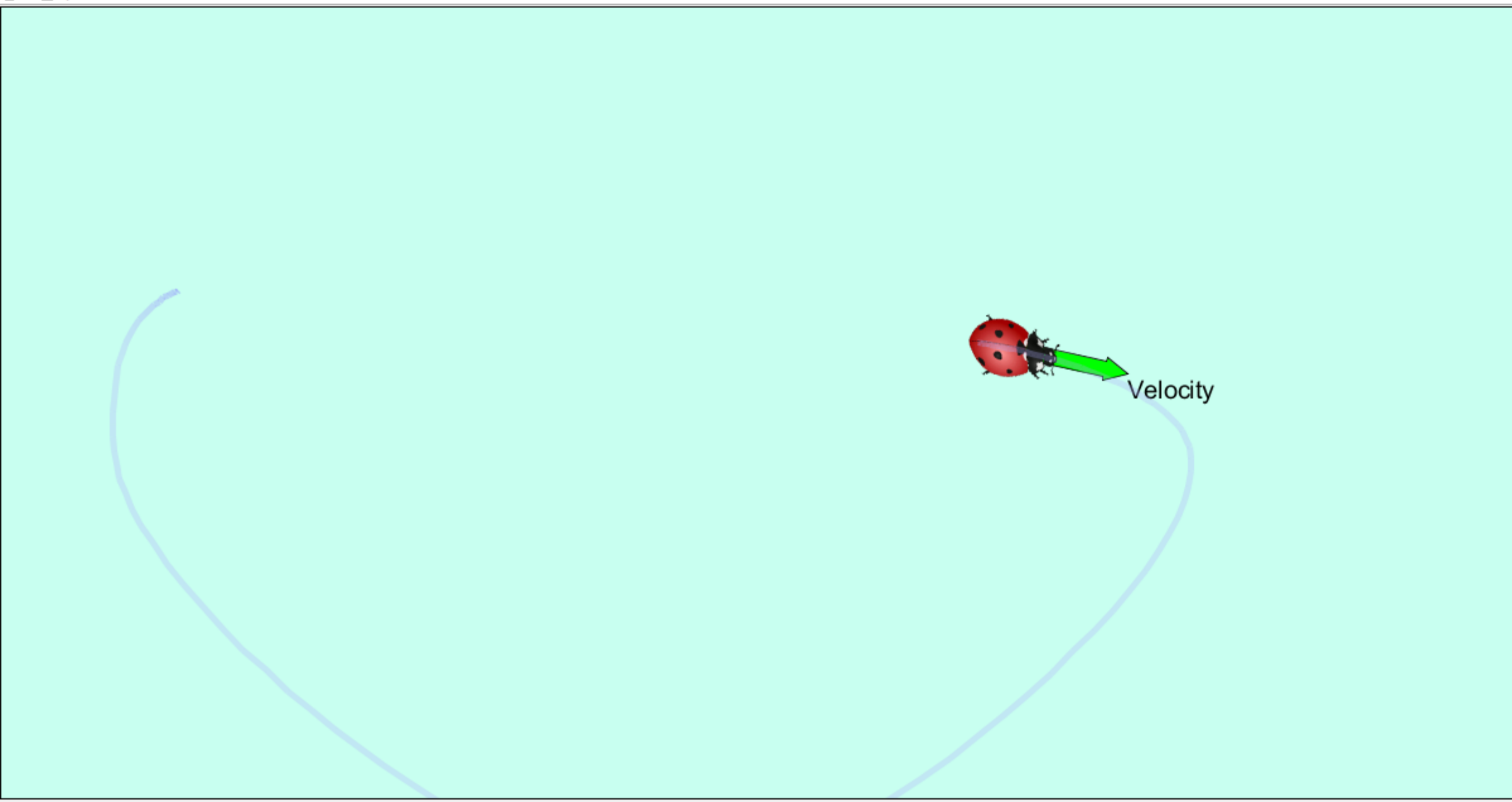
$$y = \frac{0}{6}x + 0$$

Save Line Erase Lines

- ☐ Slope
- ☐ $y = x$
- ☐ $y = -x$
- ☐ Hide lines
- ☐ Hide grid



Trajectory – Velocity Vector

**Vectors**

- ☒ Show velocity
- ☐ Show acceleration
- ☐ Show both
- ☐ Hide Vectors

Choose Motion

- ☒ Manual
- ☐ Linear
- ☐ Circular
- ☐ Ellipse

Trace

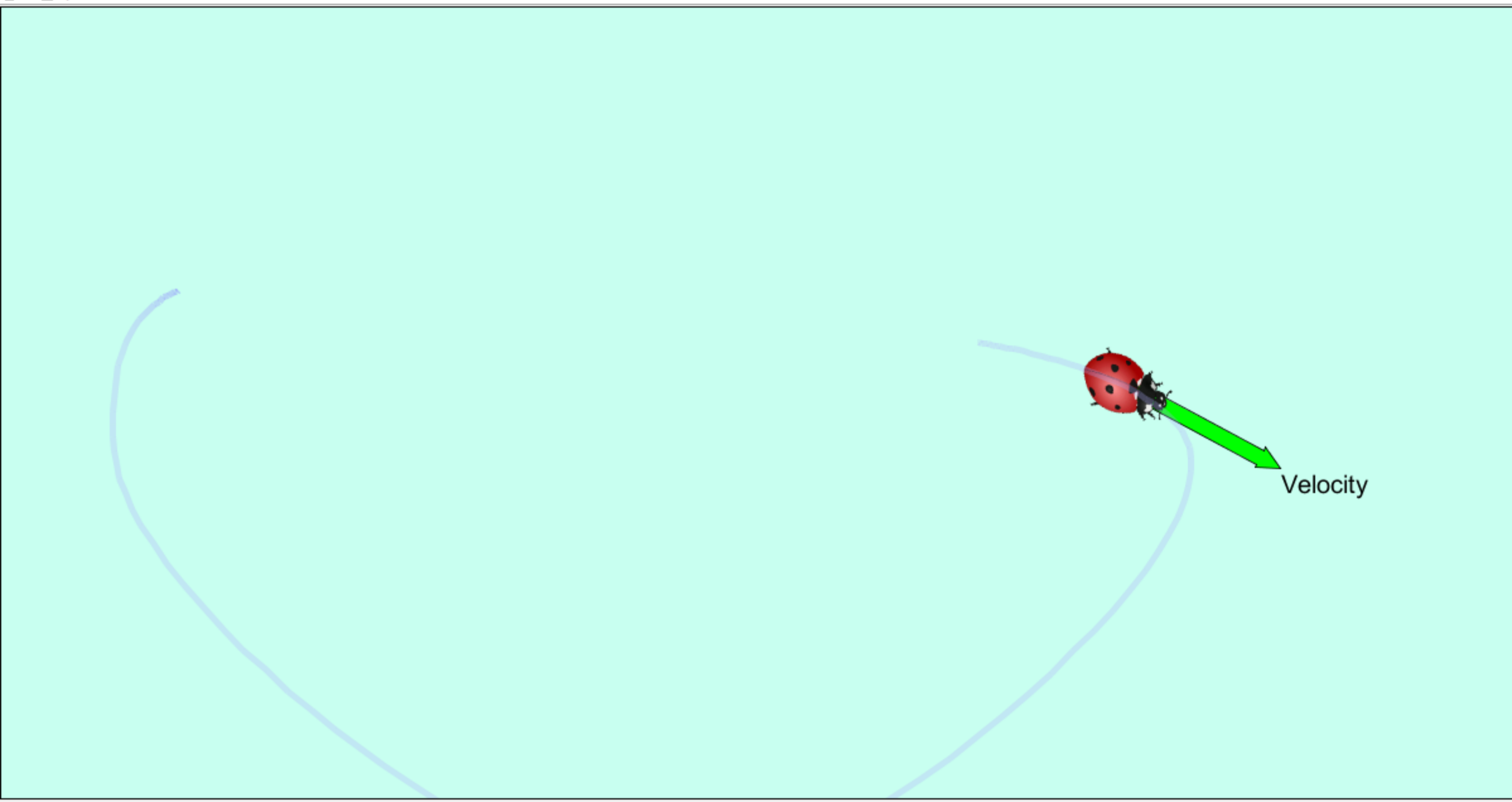
- ☒ Line
- ☐ Dots
- ☐ Off

Remote Control

- ☒ Position
- ☐ Velocity
- ☐ Acceleration

☐ Record

Res

**Vectors**

- ☒ Show velocity
- ☐ Show acceleration
- ☐ Show both
- ☐ Hide Vectors

Choose Motion

- ☒ Manual
- ☐ Linear
- ☐ Circular
- ☐ Ellipse

Trace

- ☒ Line
- ☐ Dots
- ☐ Off

Remote Control

Remote Control interface area.

- ☒ Position
- ☐ Velocity
- ☐ Acceleration

☐ Record

**Vectors**

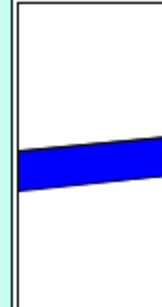
- ☒ Show velocity
- ☐ Show acceleration
- ☐ Show both
- ☐ Hide Vectors

Choose Motion

- ☒ Manual
- ☐ Linear
- ☐ Circular
- ☐ Ellipse

Trace

- ☒ Line
- ☐ Dots
- ☐ Off

Remote Control

- ☒ Position
- ☐ Velocity
- ☐ Acceleration

☐ Record



Figure 15.2. Recording of the motion of the puck with $\tau = 60$ ms.

Simulation of uniform rectilinear motion (URM)

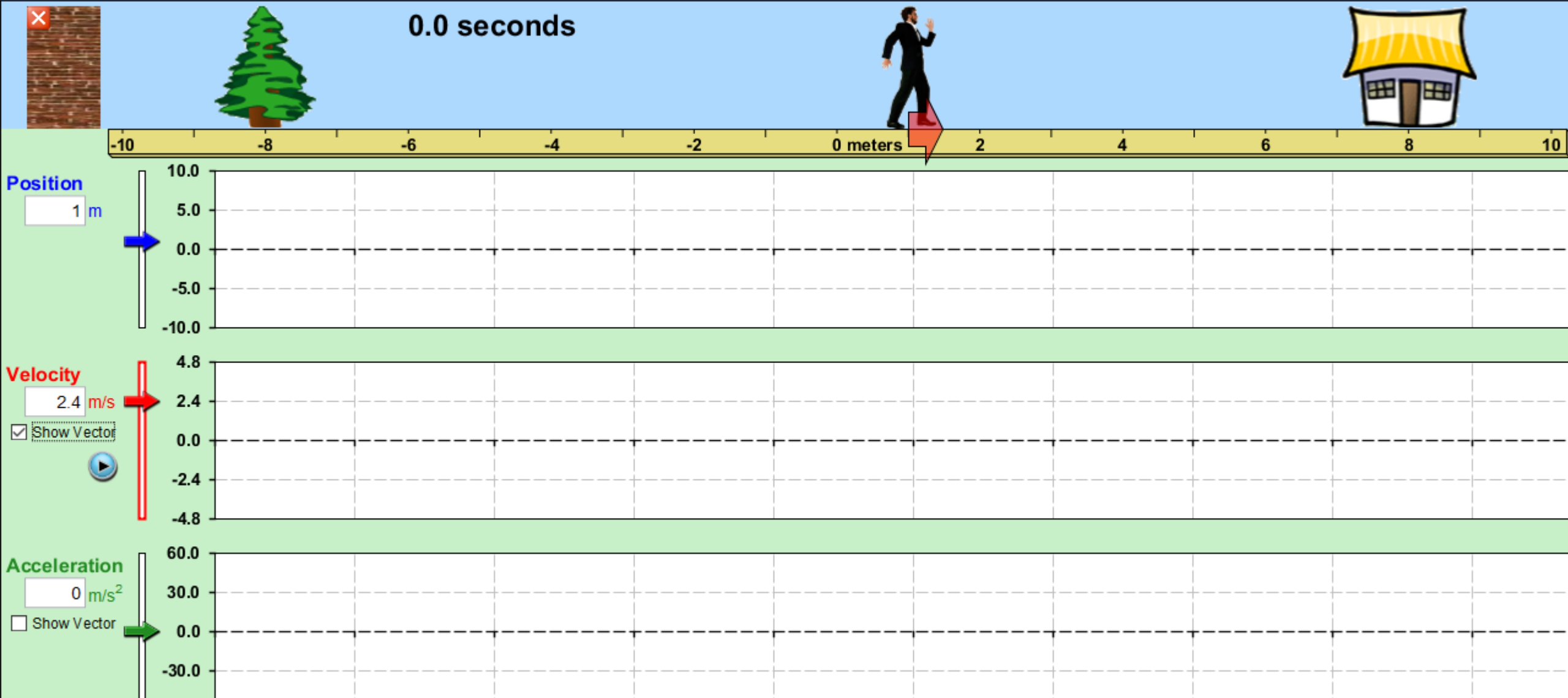
Simulation of URM with $x_0 = 1\text{m}$ and $v = 2.4\text{ m/s}$

The Moving Man (2.05)

File Special Features Help

Introduction

Charts



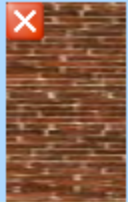
Simulation of URM with $x_0 = 1\text{m}$ and $v = 2.4\text{ m/s}$

The Moving Man (2.05)

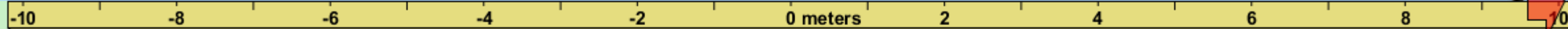
File Special Features Help

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Charts

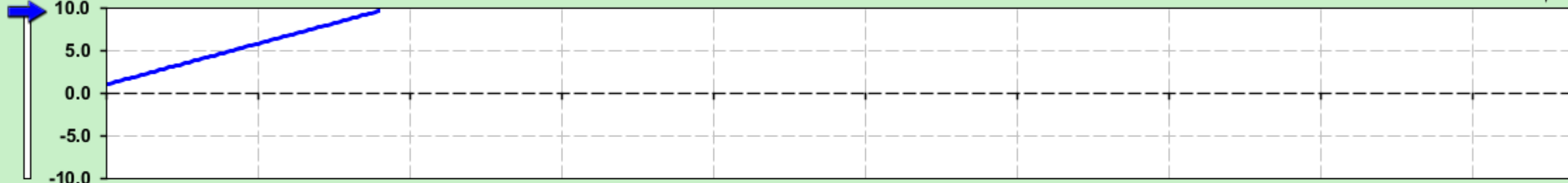


3.6 seconds



Position

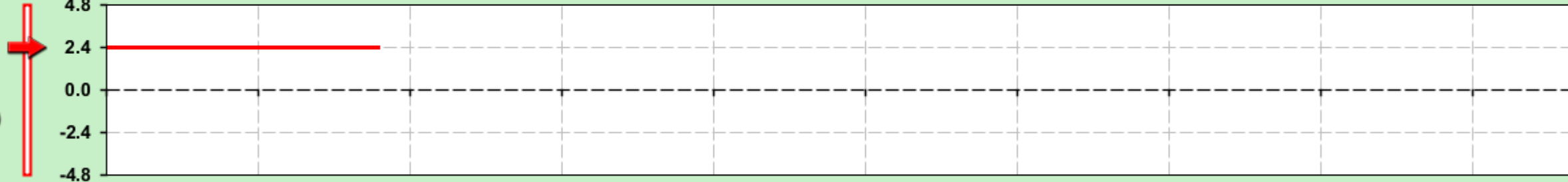
9.6 m



Velocity

2.4 m/s

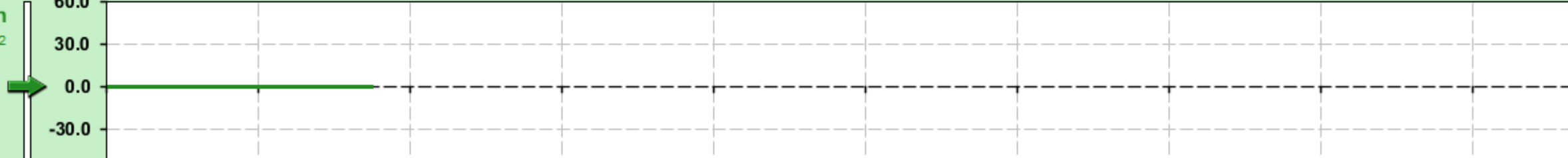
☒ Show Vector



Acceleration

0 m/s²

☐ Show Vector



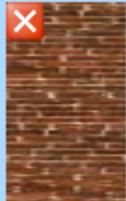
Simulation of URM with $x_0 = 1\text{m}$ and $v = 4.8\text{ m/s}$

The Moving Man (2.05)

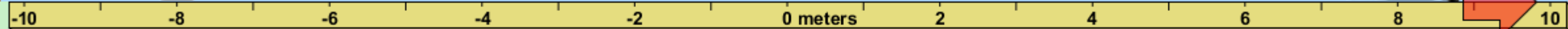
File Special Features Help

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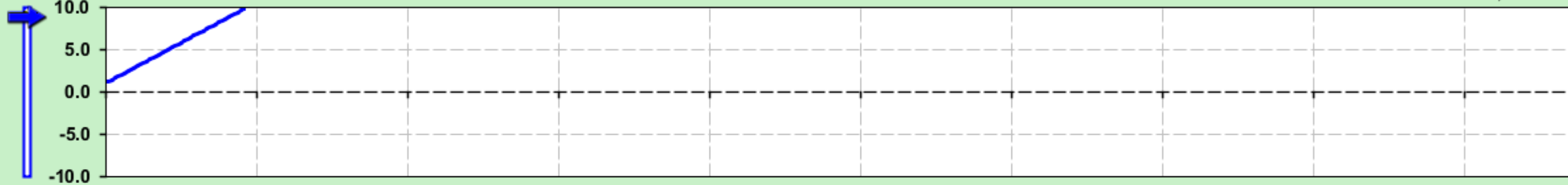


1.8 seconds



Position

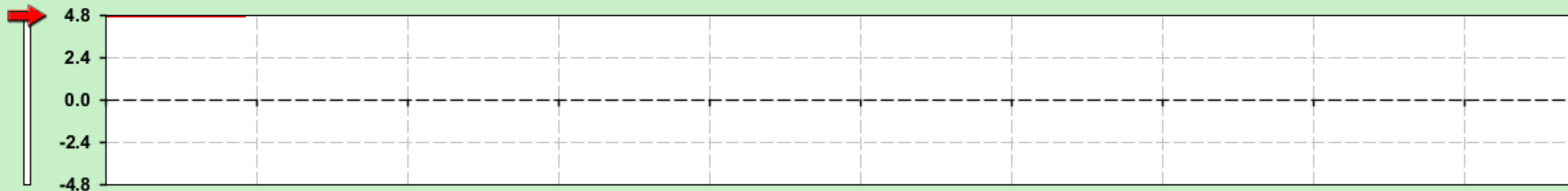
8.862 m



Velocity

4.8 m/s

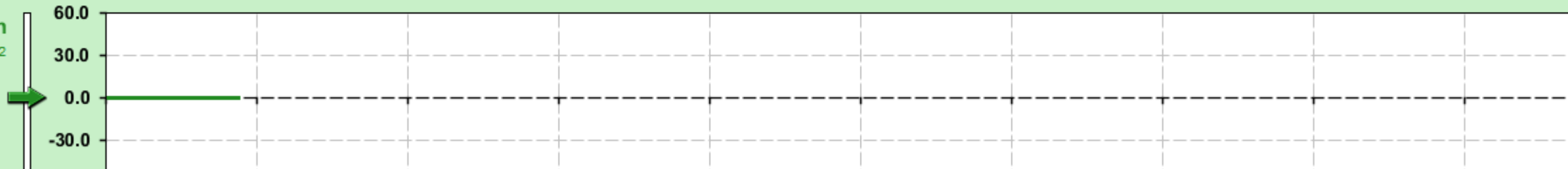
☒ Show Vector



Acceleration

0 m/s²

☐ Show Vector



Question 15.1

What is the value of the acceleration of an object moving at a uniform rectilinear motion?

Equations of URM

General Form ($x_0 \neq 0$)

- $a = 0$
- $v = \text{constante} = v_m$
- $x = v \cdot t + x_0$

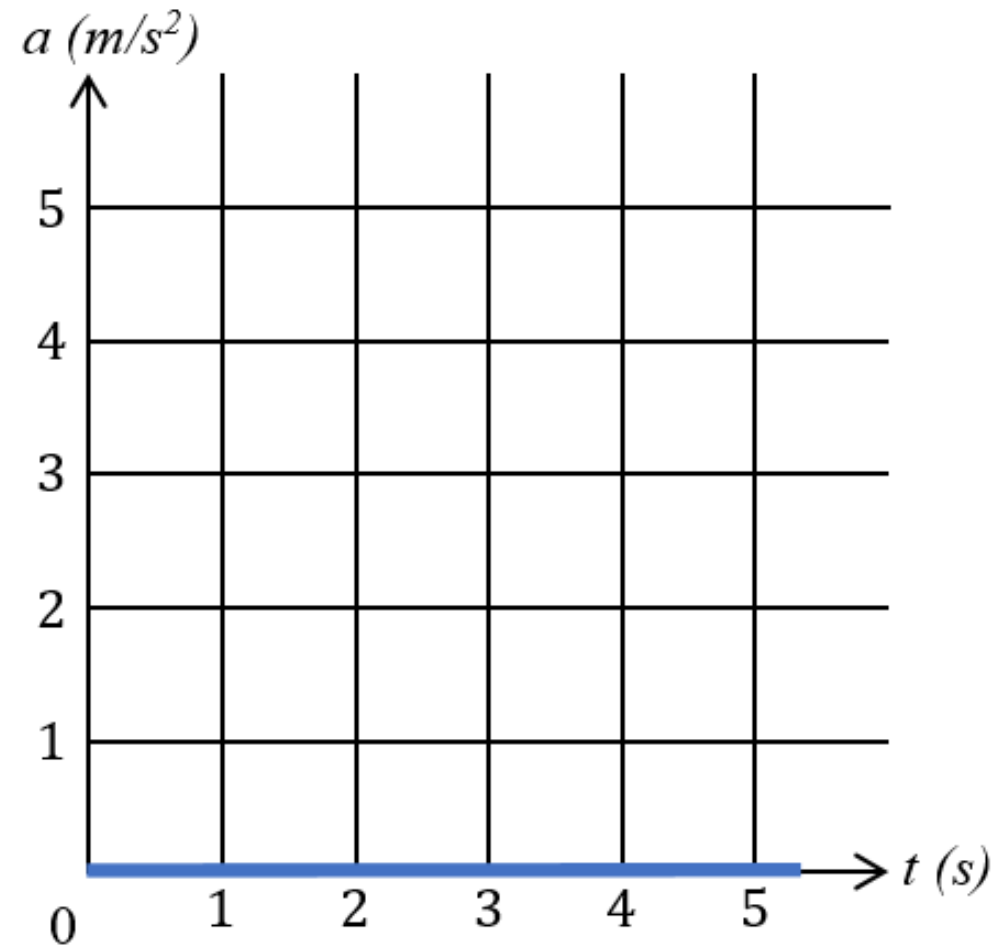
Case of an object starting from origin O ($x_0 = 0$)

- $a = 0$
- $v = \text{constante} = v_m$
- $x = v \cdot t$

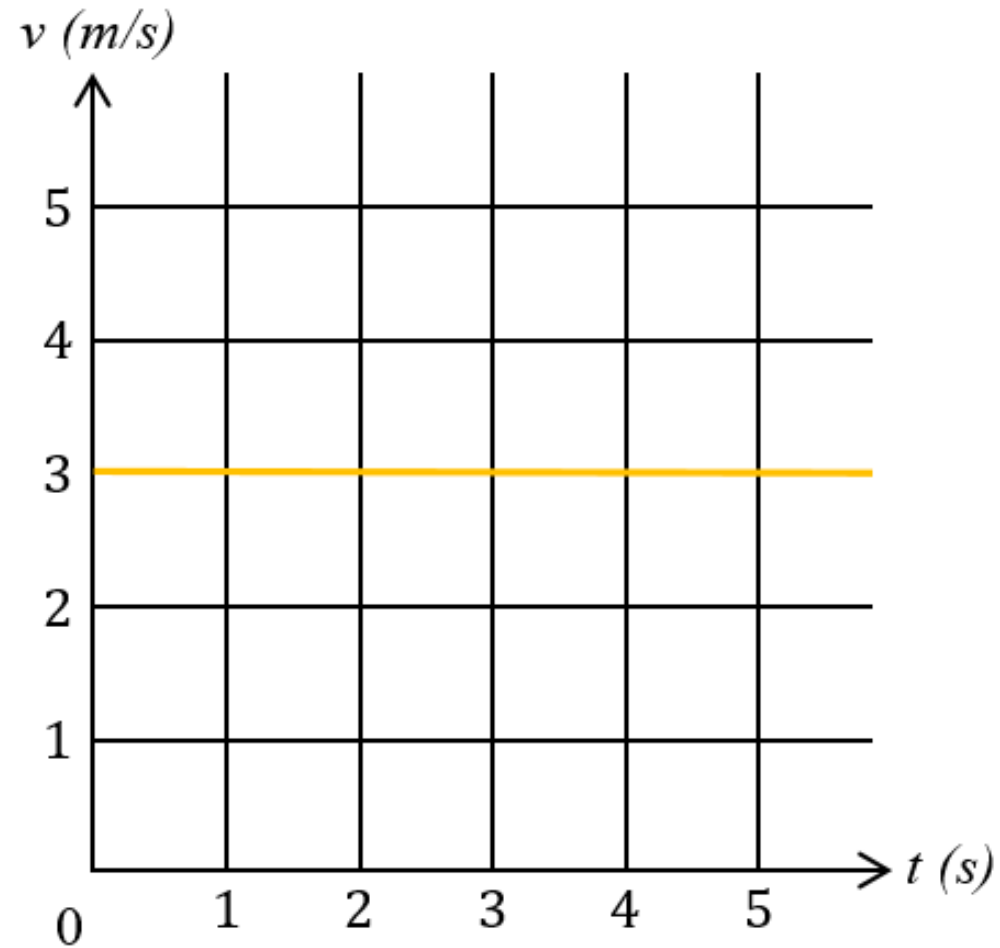
Time equation of mvt

Graphs of motion quantities as function of time

Graph of acceleration of URM ($a = 0$)

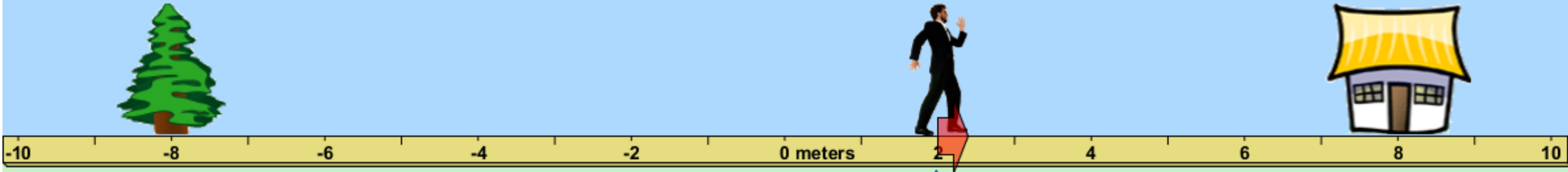


Graph of speed for URM ($v = \text{constant}$)



Graph of position vs time for URM with $x_0 \neq 0$

0.0 seconds

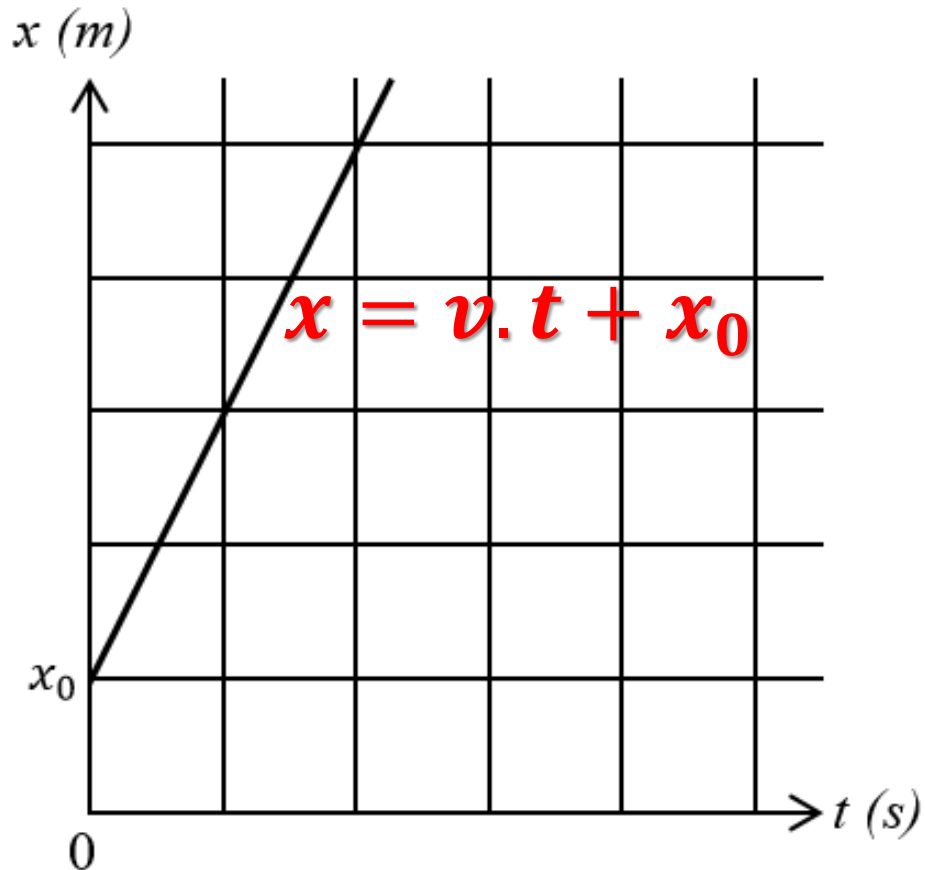


$$x_0 = 2 \text{ m}$$

Slope = velocity

$$x = v \cdot t + x_0$$

(Linear relationship – Straight line graph)



Simulation of URM with $x_0 = 0$ and $v = 2.4 \text{ m/s}$

The Moving Man (2.05)

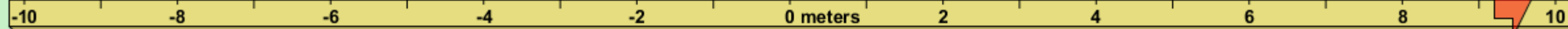
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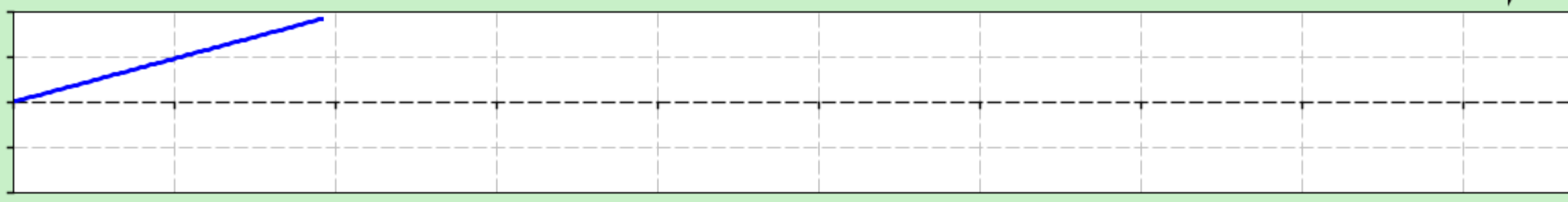
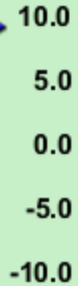


3.8 seconds



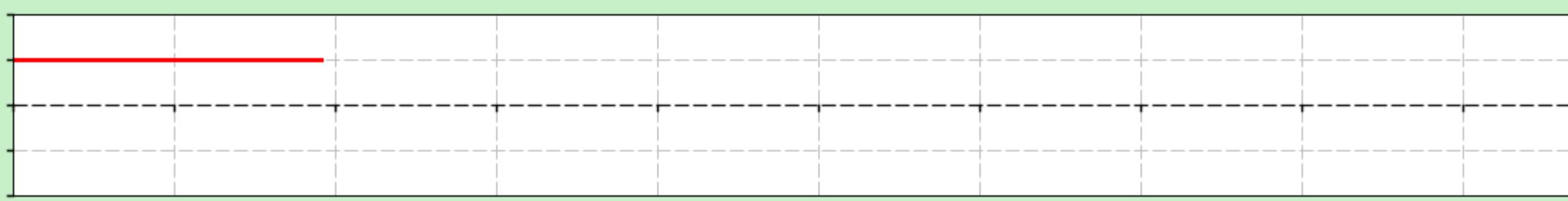
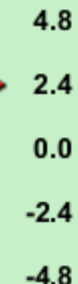
Position

9.2 m



Velocity

2.4 m/s

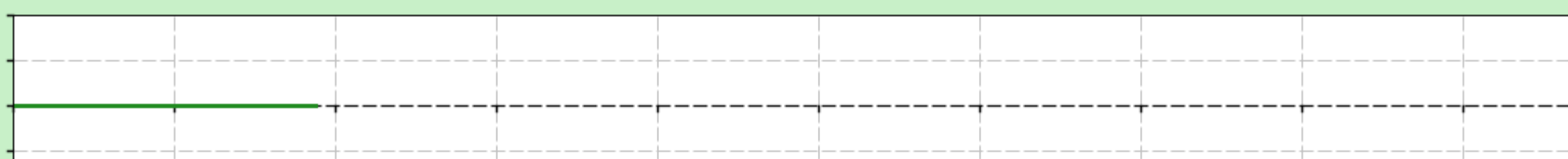
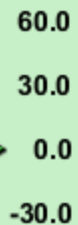


☒ Show Vector



Acceleration

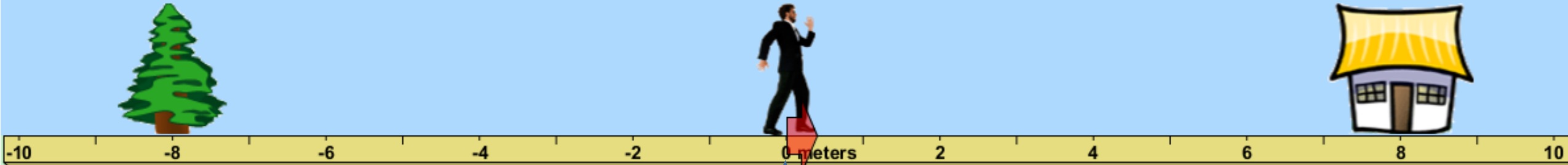
0 m/s²



☐ Show Vector

Graph of position vs time for URM starting from origin O

0.0 seconds

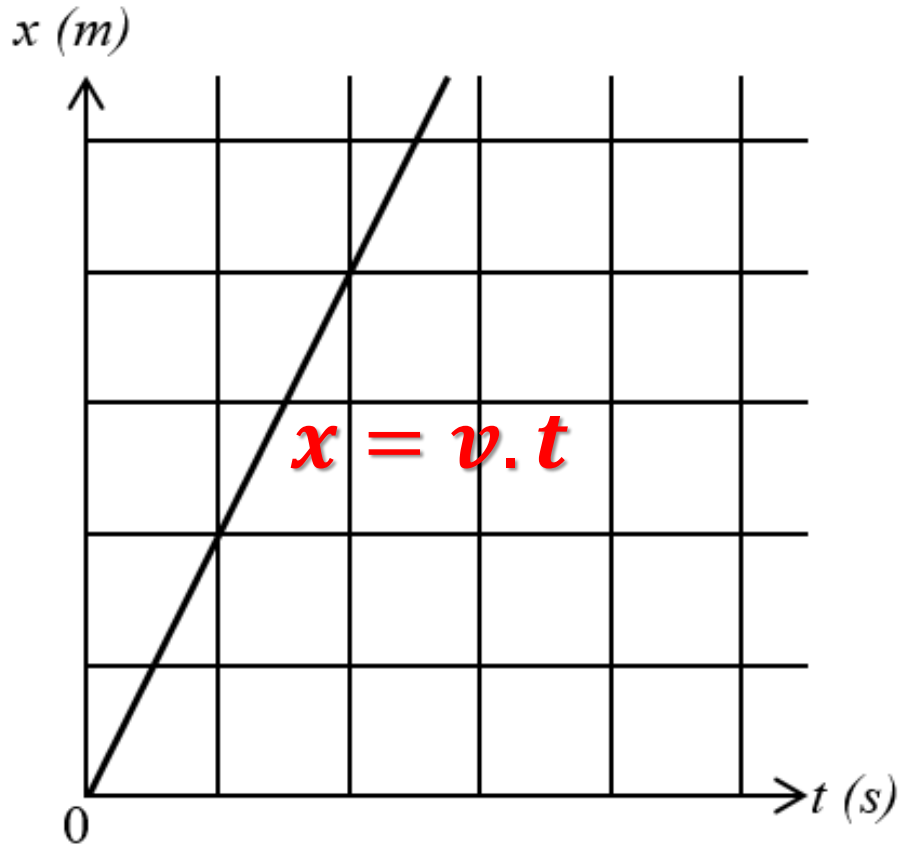


$$x_0 = 0 \text{ m}$$

$$x = v \cdot t$$

$$x = v \cdot t$$

(x proportional to time)



TEST YOUR KNOWLEDGE

1. What is a uniform rectilinear motion?
2. True or False. Correct the wrong statements.
In a uniform rectilinear motion.
 - a. The acceleration is constant.
 - b. The speed of a moving object is constant.
 - c. The average speed between any two instants is equal to the instantaneous speed.
 - d. The velocity vector of a moving object changes its direction during the motion.

3. An object, moving with a uniform rectilinear motion, has a constant speed $v = 5\text{ m/s}$. Determine its time equation if, at $t = 0$, its abscissa is 1 m .

4. A car moves with a constant speed of 25 m/s on a rectilinear track. What is the distance traveled in 10 minutes?

QUESTIONS

1. The graphs of figure 15.9 represent rectilinear motions. Which of them corresponds to a uniform rectilinear motion?

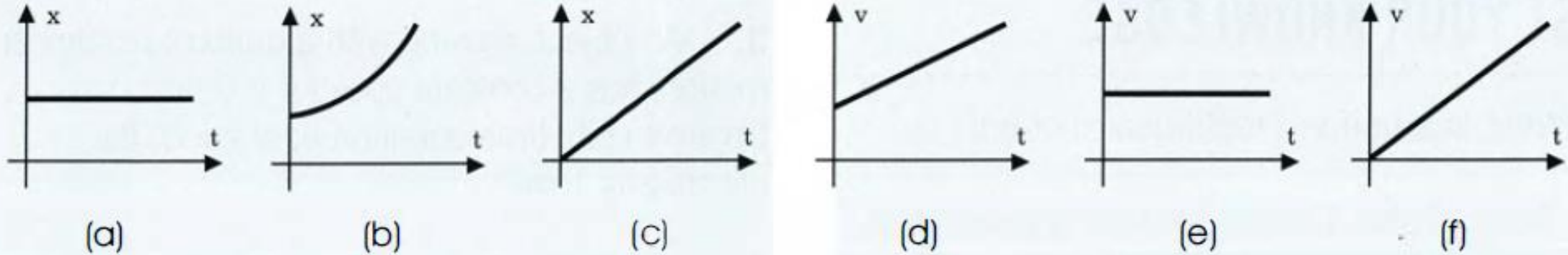


Figure 15.9.

PROBLEMS

1. The time equation of a body in rectilinear motion is $x = 10t + 2$ in SI units.

- What is the nature of its motion?
- Determine the positions of the body at $t = 0$ and at $t = 1$ min.
- What is the direction of motion on the $x - \text{axis}$?
- What is the acceleration vector?

2. A car moves at a constant speed of 80 km/h on a rectilinear path. How long does it take to travel 100 km?

7 Figure 15.11 shows the recording of the motion of the center C of a puck thrown on a horizontal air table. The marks on the paper are separated by equal time intervals $t = 1/50$ s. The successive positions of C are: $C_0(t_0)$, $C_1(t_1)$, $C_2(t_2)$, ..., $C_i(t_i)$..., where t_0 is taken as origin of time.

a. Compare the distances traveled by the puck in equal time intervals. Conclude.

- b. Calculate the average speed of the puck between any two instants you choose. Conclude.
- c. Calculate the instantaneous speed of the puck at any three instants you choose. Conclude.
- d. Calculate the average acceleration of the puck between any two instants you choose. Conclude.
- e. What is the nature of the motion?

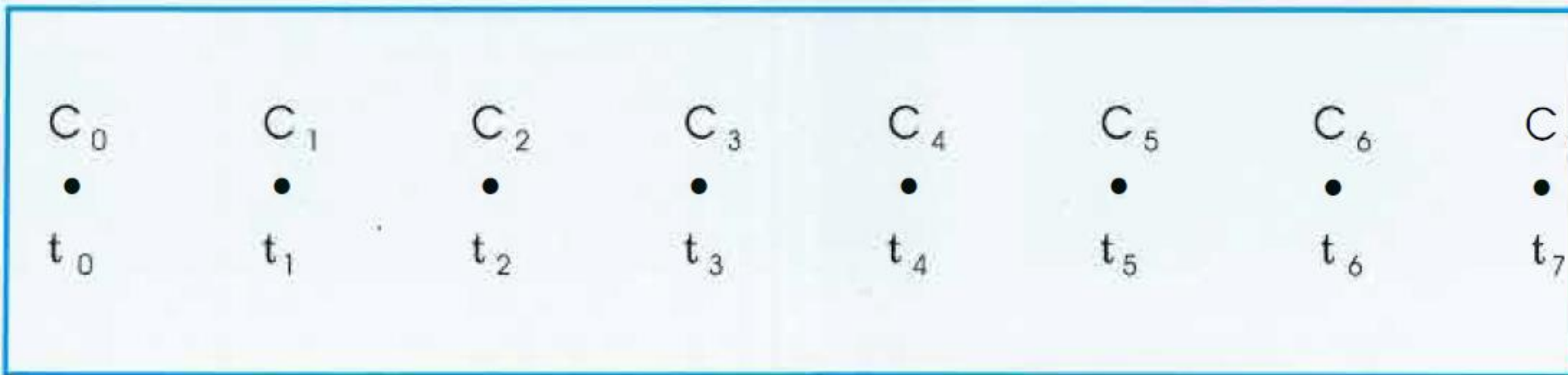


Figure 15.11.